

INTRODUCTION TO PHYSICAL SCIENCE



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Introduction to Physical Science

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Licensing

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CHAPTER OVERVIEW

1: What is Science?

Topic hierarchy

- [1.1: Lecture 1 - What is Science?](#)
- [1.2: Lecture 2 - Physical Quantities and Units](#)
- [1.3: Lecture 3 - Accuracy, Precision, and Significant Figures](#)

This lesson is based on Chapter 1 of the OpenStax College Physics textbook. Each section of the chapter will include my lecture slides as well as an embedded link to my YouTube video for that lecture.

1: [What is Science?](#) is shared under a [CC BY 4.0](#) license and was authored, remixed, and/or curated by LibreTexts.

1.1: Lecture 1 - What is Science?

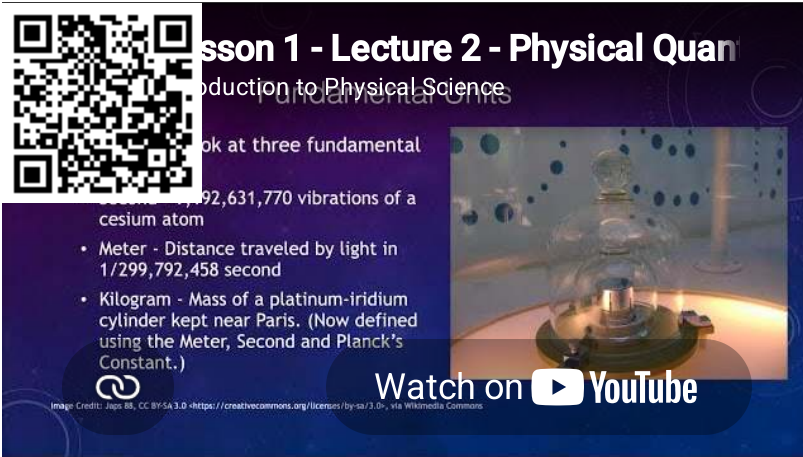
The lesson is based on section 1.1 in the OpenStax College Physics textbook. The lecture slides are provided in [PowerPoint](#), [Keynote](#), and [pdf](#) format.



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1.2: Lecture 2 - Physical Quantities and Units

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Lesson 1 - Lecture 2 - Physical Quantities and Units
Introduction to Physical Science

Look at three fundamental units:

- Second - 919,263,170 vibrations of a cesium atom
- Meter - Distance traveled by light in $1/299,792,458$ second
- Kilogram - Mass of a platinum-iridium cylinder kept near Paris. (Now defined using the Meter, Second and Planck's Constant.)

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1.3: Lecture 3 - Accuracy, Precision, and Significant Figures

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Lesson 1 - Lecture 3 - Accuracy, Precision, and Significant Figures
Introduction to Physical Science

Leading zeros are ALWAYS significant

- Leading zeros are NEVER significant
- Embedded zeros (zeros appearing between two non-zero digits) are significant
- Trailing zeros are significant ONLY if the decimal point is specified
- All digits to the left of the $\times 10^x$ in scientific notation are significant

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CHAPTER OVERVIEW

2: Motion in One Dimension

Topic hierarchy

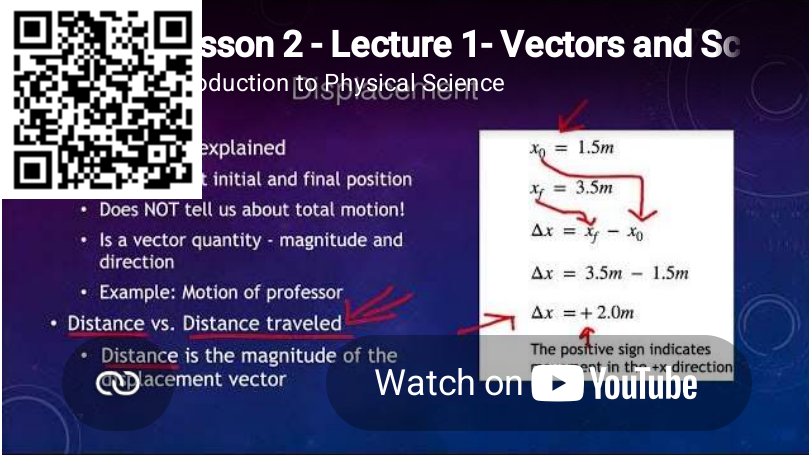
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- [2.5: Lecture 5 - Newton's Laws and Gravity](#)

This lesson is based on Chapter 2 (2.1-2.7) and Chapter 4 (4.1-4.4) and Chapter 6 (6.5) of the OpenStax College Physics textbook. Each section of the chapter will include my lecture slides as well as an embedded link to my YouTube video for that lecture.

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2.1: Lecture 1 - Vectors and Scalars

The lesson is based on sections 2.1-2.2 in the OpenStax College Physics textbook. The lecture slides are provided in [PowerPoint](#), [Keynote](#), and [pdf](#) format.



Lesson 2 - Lecture 1- Vectors and Scalars
Introduction to Physical Science

Displacement explained

- Does NOT tell us about total motion!
- Is a vector quantity - magnitude and direction
- Example: Motion of professor
- **Distance vs. Distance traveled**
- Distance is the magnitude of the displacement vector

$x_0 = 1.5m$
 $x_f = 3.5m$
 $\Delta x = x_f - x_0$
 $\Delta x = 3.5m - 1.5m$
 $\Delta x = +2.0m$

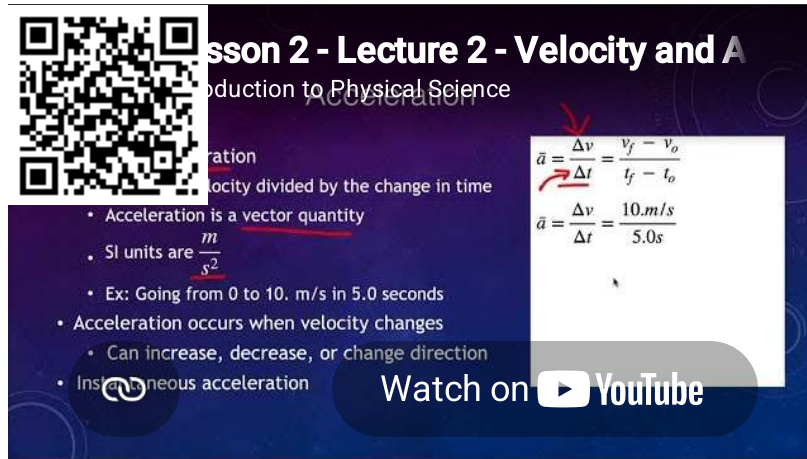
The positive sign indicates displacement in the +x direction

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2.2: Lecture 2 - Velocity and Acceleration

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The slide features a QR code in the top left corner. The title is "Lesson 2 - Velocity and Acceleration" with a subtitle "Introduction to Physical Science". The main text defines acceleration as "velocity divided by the change in time". A list of bullet points includes: "Acceleration is a vector quantity", "SI units are $\frac{m}{s^2}$ ", "Ex: Going from 0 to 10. m/s in 5.0 seconds", "Acceleration occurs when velocity changes" (with sub-points "Can increase, decrease, or change direction"), and "Instantaneous acceleration". A callout box on the right shows the formula $\vec{a} = \frac{\Delta v}{\Delta t} = \frac{v_f - v_o}{t_f - t_o}$ and a calculation $\vec{a} = \frac{\Delta v}{\Delta t} = \frac{10.m/s}{5.0s}$. A "Watch on YouTube" button is at the bottom right.

Lesson 2 - Velocity and Acceleration
Introduction to Physical Science

Acceleration
velocity divided by the change in time

- Acceleration is a vector quantity
- SI units are $\frac{m}{s^2}$
- Ex: Going from 0 to 10. m/s in 5.0 seconds
- Acceleration occurs when velocity changes
 - Can increase, decrease, or change direction
- Instantaneous acceleration

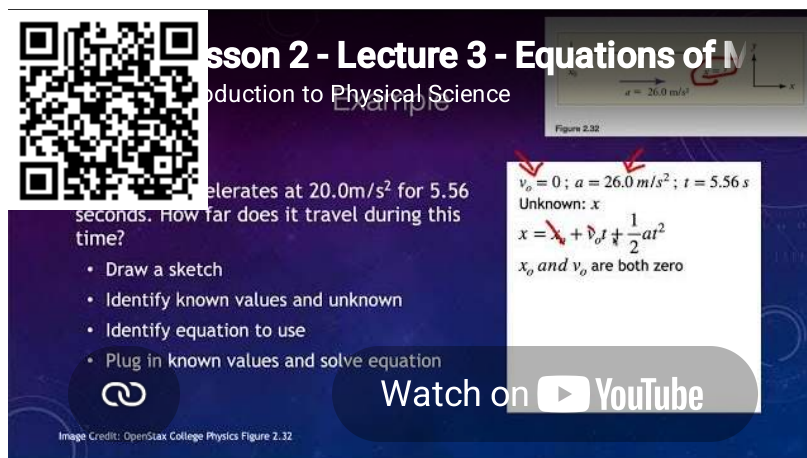
$$\vec{a} = \frac{\Delta v}{\Delta t} = \frac{v_f - v_o}{t_f - t_o}$$
$$\vec{a} = \frac{\Delta v}{\Delta t} = \frac{10.m/s}{5.0s}$$

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2.3: Lecture 3 - Equations of Motion

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Lesson 2 - Lecture 3 - Equations of Motion
Introduction to Physical Science

Example
An object starts from rest and accelerates at 20.0 m/s^2 for 5.56 seconds. How far does it travel during this time?

- Draw a sketch
- Identify known values and unknown
- Identify equation to use
- Plug in known values and solve equation

Image Credit: OpenStax College Physics Figure 2.32

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Figure 2.32
 $x = 26.0 \text{ m/s}^2$

$v_o = 0$; $a = 26.0 \text{ m/s}^2$; $t = 5.56 \text{ s}$
Unknown: x
 $x = v_o t + \frac{1}{2} a t^2$
 x_o and v_o are both zero

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